STAMPose: Spatial-Temporal Attention-Mamba for Efficient 3D Human Pose Estimation

Hyun-il Kim[1] Seung-bo Park[2]
[1]Dept. of Advanced Material Processing & Engineering, Inha University [2]Dept. of Software Convergence Engineering, Inha University, Incheon, Korea
akadjsam@inha.edu | molaal@inha.ac.kr

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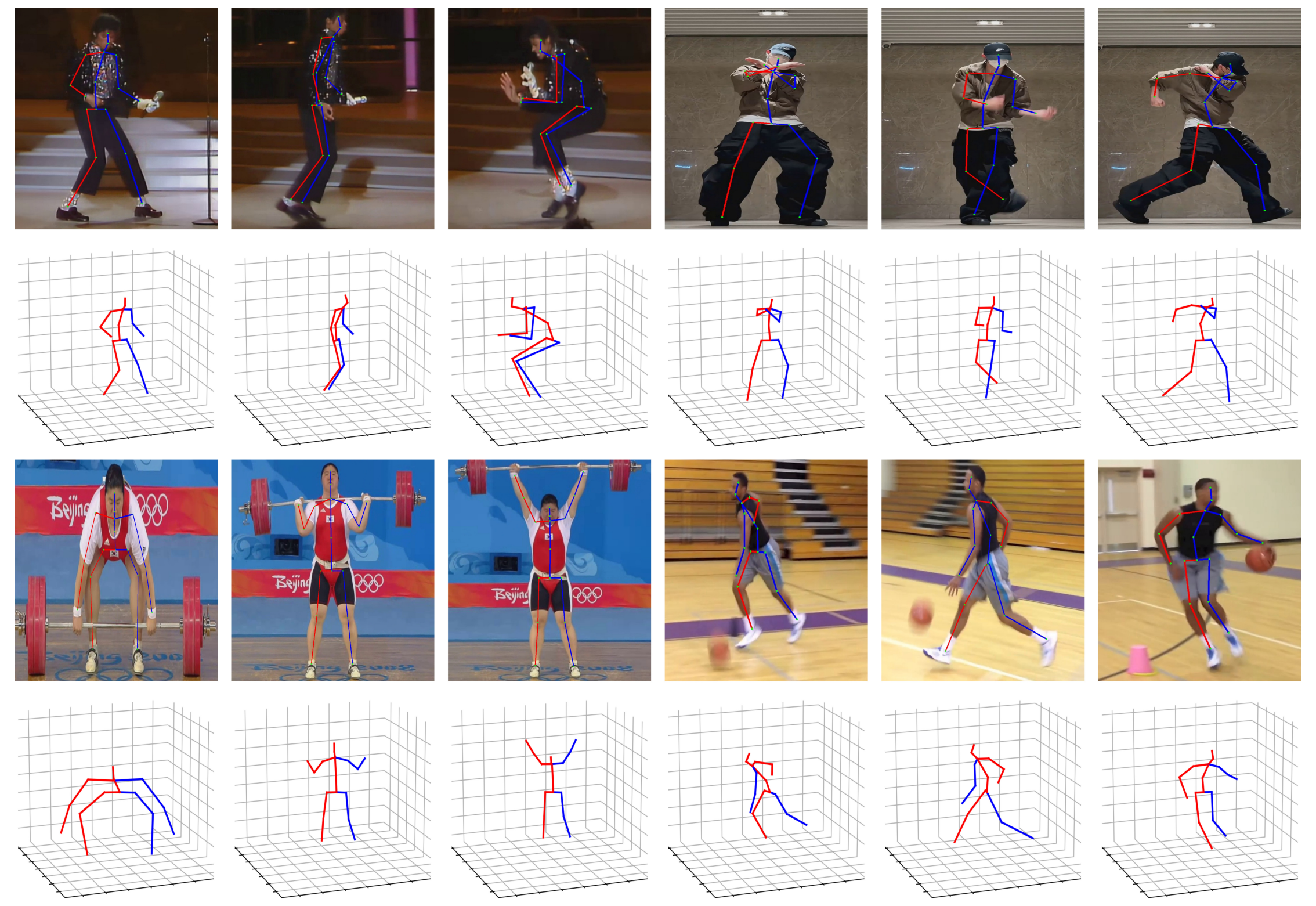
Code and models

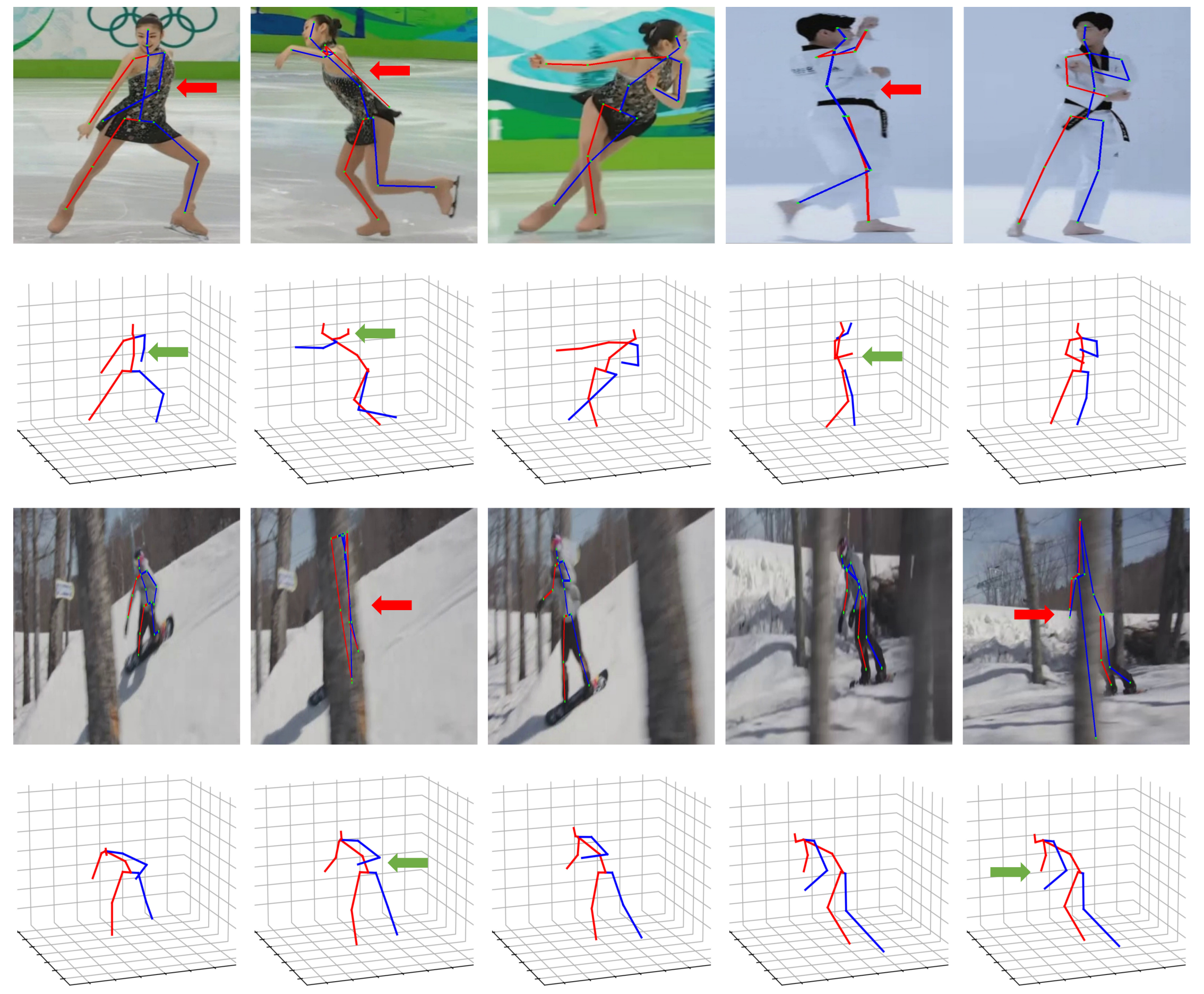
Contributions

1. Attention on the spatial axis and a Mamba block on the temporal axis, chosen by measurement.
2. DC-TSSM concatenates a bidirectional scan branch with a depthwise convolution branch.
3. 1.7–2.0× faster than PoseMamba at comparable size, with comparable or better accuracy.

STAMPose in the Wild

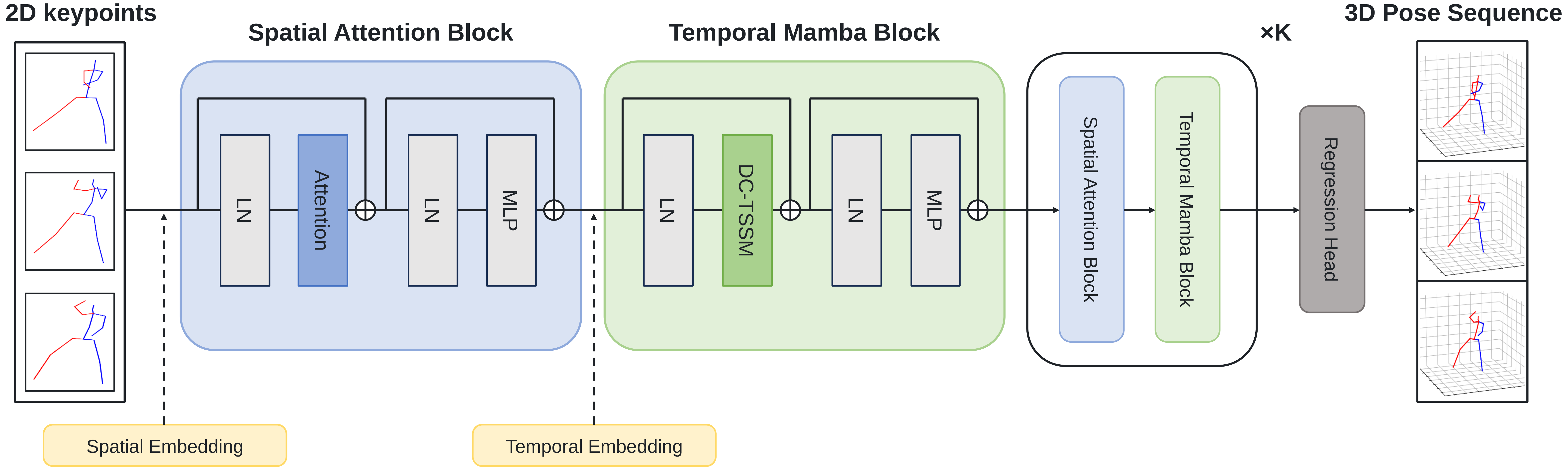


STAMPose-L on unseen videos. Rows 1 and 3: inputs with 2D keypoints; rows 2 and 4: predicted 3D poses.



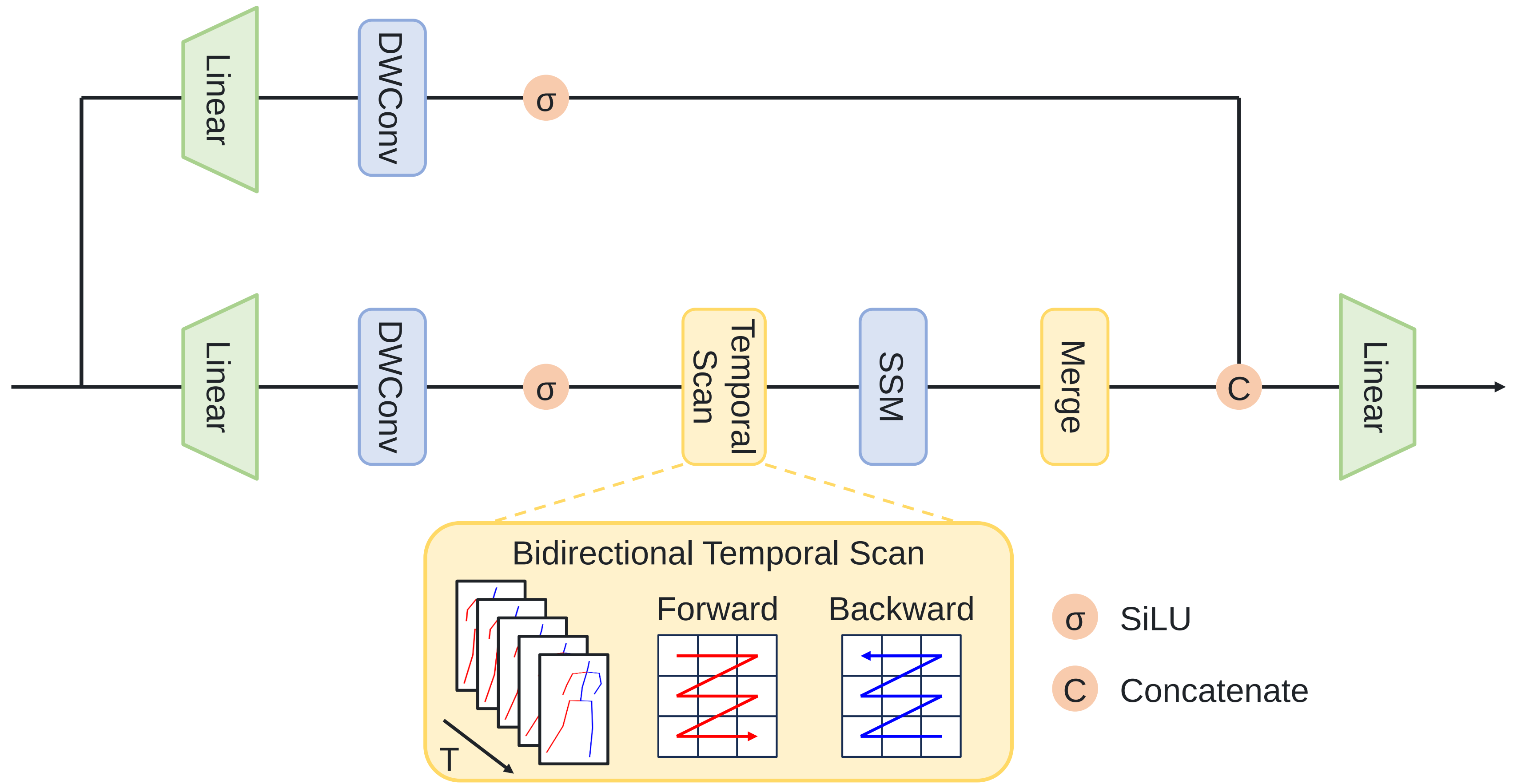
Clips where the 2D detector fails. Red arrows: wrong 2D keypoints; green arrows: coherent 3D poses.

STAMPose Architecture



2D keypoints pass through K pairs of a **Spatial Attention Block** (S) and a **Temporal Mamba Block** (T), stacked as (ST)^K.

Dual-Branch Concatenation Temporal SSM



DC-TSSM concatenates its depthwise-conv branch with the bidirectional scan branch instead of using it as a multiplicative gate.

Comparison with the State of the Art

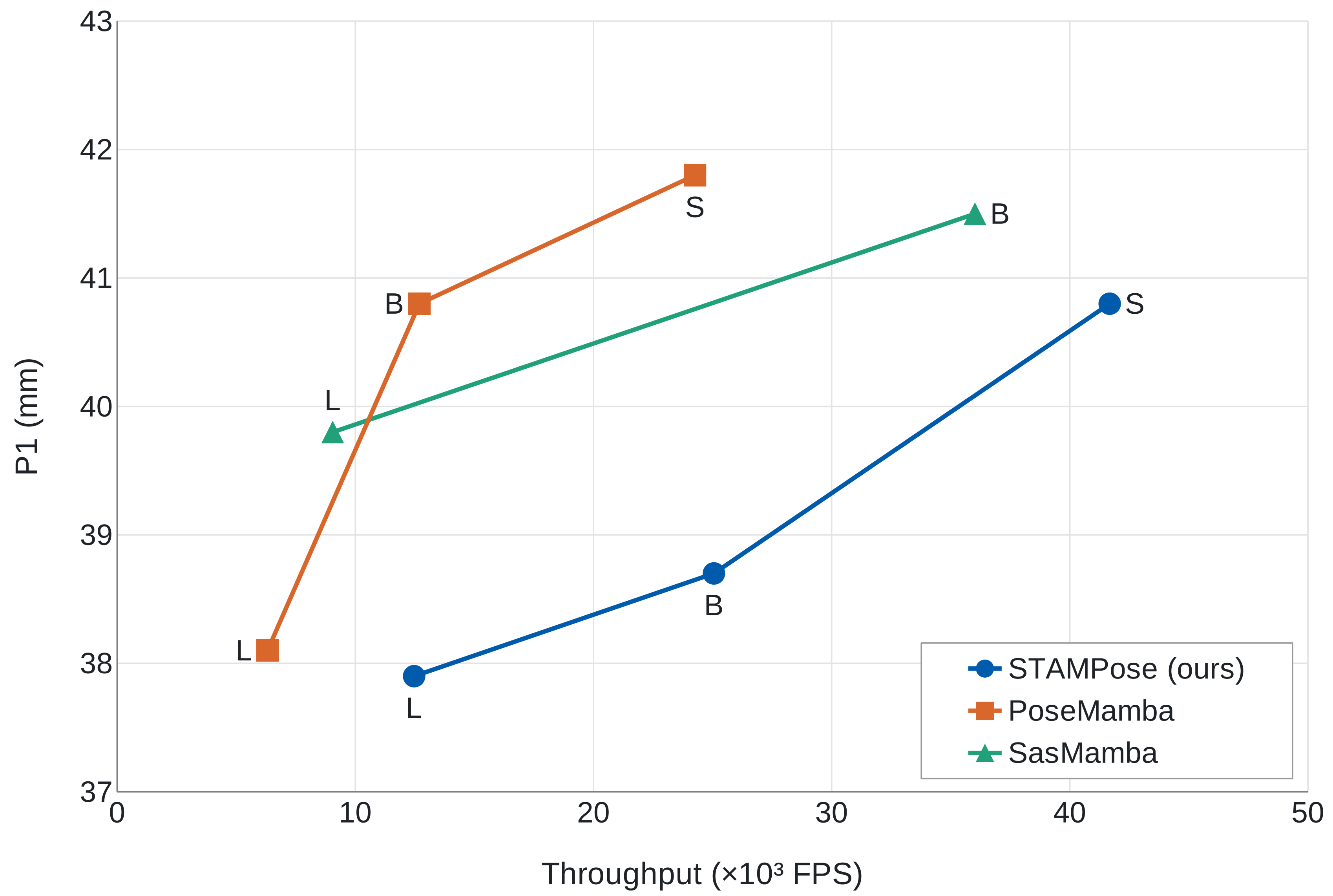
Method	Params	MACs/frame	P1 ↓	P2 ↓	P1† ↓
MotionBERT	42.3M	719M	39.2	32.9	17.8
MotionAGFormer-L	19.0M	322M	38.4	32.5	17.3
KTPFormer	33.7M	286M	40.1	31.9	19.0
TCPFormer	35.1M	449M	**37.9**	**31.7**	15.5
HGMamba	14.2M	265M	38.7	32.9	13.2
PoseMamba-S	0.9M	15M	41.8	35.0	20.0
PoseMamba-B	3.4M	57M	40.8	34.3	16.8
PoseMamba-L	6.7M	115M	38.1	32.5	15.6
SasMamba	0.64M	5M	41.5	34.8	21.4
SasMamba-large	4.1M	35M	39.8	33.6	20.9
STAMPose-S	0.8M	14M	40.8	34.3	18.6
STAMPose-B	3.3M	57M	38.7	33.0	14.0
STAMPose-L	6.7M	113M	**37.9**	32.2	**12.3**

Human3.6M (T = 243). P1 / P2: MPJPE / P-MPJPE (mm); P1†: ground-truth 2D input. Bold: best, underline: second best.

Method	T	PCK ↑	AUC ↑	MPJPE ↓
MotionAGFormer-L	81	98.2	85.3	16.2
KTPFormer	81	98.9	85.9	16.7
TCPFormer	81	99.0	87.7	15.0
PoseMamba	81	–	–	14.5
SasMamba	81	–	–	18.0
STAMPose	81	**99.6**	**89.2**	**14.2**

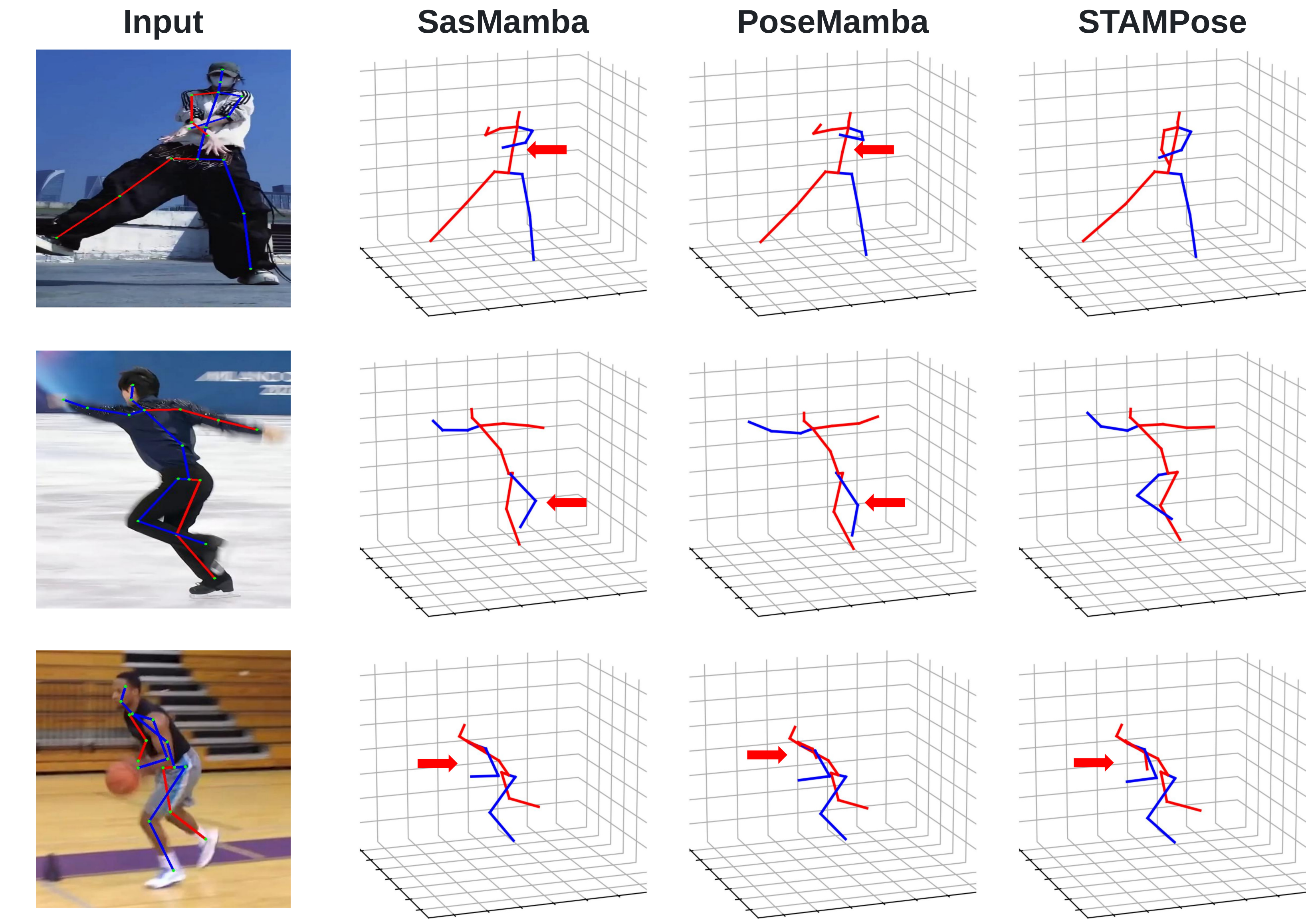
MPI-INF-3DHP (T = 81). –: not reported.

Inference Efficiency



Accuracy vs. throughput on Human3.6M. STAMPose is 1.72×, 1.97× and 1.98× faster than PoseMamba-S, -B and -L. FPS: one RTX 3090 Ti, batch 32, T = 243.

Qualitative Comparison



In-the-wild comparison at similar size: SasMamba-large (4.1M), PoseMamba-B (3.4M) and STAMPose-B (3.3M). Red arrows: incorrect joints.